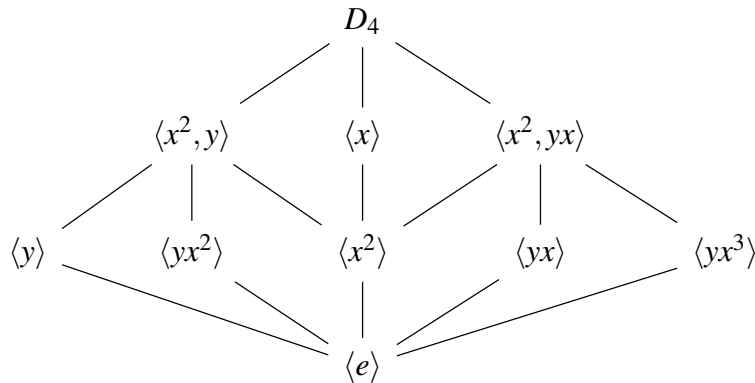


HW 1

Problem 1

a)



Let H be a subgroup of D_4 by Lagrange's Theorem the order of H can only be 1, 2, 4, 8

Case 1: H has order 1

- The only subgroup of order 1 is the trivial subgroup $\{e\}$.
- Since $\{e\}$ is already on our list, H cannot be of order 1 unless it is $\{e\}$, which is already accounted for.

Case 2: H has order 2

- If H has order 2, it must consist of e and one other element g , where $g^2 = e$.
- In D_4 , the elements that satisfy $g^2 = e$ are x^2 , y , yx , yx^2 , and yx^3 .
- Each of these elements generates a subgroup of order 2:

- $\langle x^2 \rangle = \{e, x^2\}$
- $\langle y \rangle = \{e, y\}$
- $\langle yx \rangle = \{e, yx\}$
- $\langle yx^2 \rangle = \{e, yx^2\}$
- $\langle yx^3 \rangle = \{e, yx^3\}$

- Therefore, any subgroup H of order 2 must be one of these, and all are already on our list.

Case 3: H has order 4

- If H has an element of order 4 this implies $x \in H$ or $x^3 \in H$ since $\langle x \rangle = \langle x^3 \rangle$ this implies $\langle x \rangle \subset H$ but since they both have 4 elements this means $H = \langle x \rangle$

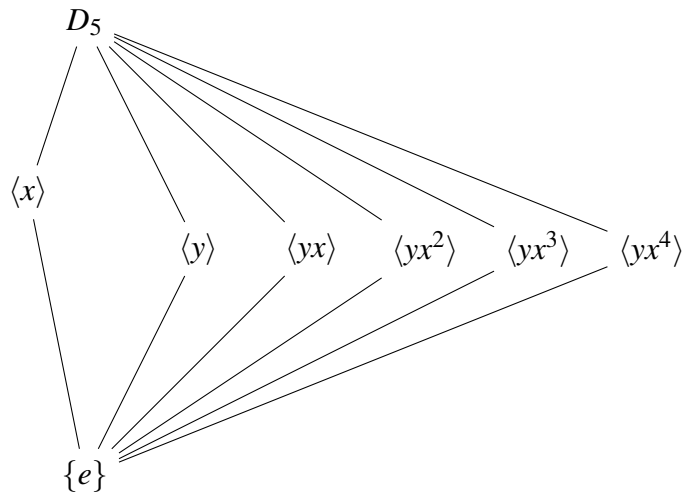
- If H does not have an element of order 4 this implies all non-identity elements must be order 2 by Lagrange's Theorem. The elements of order 2 are $\{x^2, y, yx, yx^2, yx^3\}$
 - If $yx^3 \in H$ and we then add $a \in \{x^2, y, yx, yx^2\}$
 - * If $a = x^2$ then $yx^3x^2 = x^2xy = yx \in H \implies H = \langle x^2, yx \rangle$
 - * If $a = yx$ then $yx^3yx = x^2 \in H \implies H = \langle x^2, yx \rangle$
 - * If $a = y$ then $yx^3y = x^3 \in H \implies$ is bigger than 4.
 - * If $a = yx^2$ then $yx^3yx^2 = x \in H \implies$ is bigger than 4.
 - If $yx \in H$ and we then add $a \in \{x^2, y, yx^2, yx^3\}$
 - * If $a = x^2$ then $yx^2 = yx^3 \in H \implies H = \langle x^2, yx \rangle$
 - * If $a = yx^3$ then $yx^3yx = x^2 \in H \implies H = \langle x^2, yx \rangle$
 - * If $a = y$ then $yyx = x \in H \implies$ is bigger than 4.
 - * If $a = yx^2$ then $yx^2yx = x^3 \in H \implies$ is bigger than 4.
 - If $y \in H$ and we then add $a \in \{x^2, yx, yx^2, yx^3\}$
 - * If $a = x^2$ then $yx^2 \in H \implies H = \langle x^2, y \rangle$
 - * If $a = yx^2$ then $x^2 \in H \implies H = \langle x^2, y \rangle$
 - * If $a = yx^3$ then $yyx^3 = x^3 \in H \implies$ is bigger than 4.
 - * If $a = yx$ then $yyx = x \in H \implies$ is bigger than 4.
 - If $x^2 \in H$ and we then add $a \in \{y, yx, yx^2, yx^3\}$
 - * If $a = y$ then $yx^2 \in H \implies H = \langle x^2, y \rangle$
 - * If $a = yx$ then $yx^3 \in H \implies H = \langle x^2, yx \rangle$
 - * If $a = yx^2$ then $y \in H \implies H = \langle x^2, y \rangle$
 - * If $a = yx^3$ then $yx \in H \implies H = \langle x^2, yx \rangle$

Case 4: H has order 8

- If H has order 8, it must be the whole group D_4 .
- Therefore, H would be D_4 , which is already on our list.

In each case, we've shown that any possible subgroup H of D_4 must be one of the subgroups already listed. Therefore, no additional subgroups can exist, and our list of subgroups is complete.

b)



Let H be a subgroup of D_5 by Lagrange's Theorem the order of H can only be 1, 2, 5, 10

Case 1: H has order 1

- The only subgroup of order 1 is the trivial subgroup $\{e\}$.
- Since $\{e\}$ is already on our list, H cannot be of order 1 unless it is $\{e\}$, which is already accounted for.

Case 2: H has order 2

- If H has order 2, it must consist of e and one other element g , where $g^2 = e$.
- In D_4 , the elements that satisfy $g^2 = e$ are y, yx, yx^2, yx^3 , and yx^4 .
- Each of these elements generates a subgroup of order 2:
 - $\langle y \rangle = \{e, y\}$
 - $\langle yx \rangle = \{e, yx\}$
 - $\langle yx^2 \rangle = \{e, yx^2\}$
 - $\langle yx^3 \rangle = \{e, yx^3\}$
 - $\langle yx^4 \rangle = \{e, yx^4\}$
- Therefore, any subgroup H of order 2 must be one of these, and all are already on our list.

Case 3: H has order 5

- Proven in class since 5 is prime this means H is cyclic which implies H has an element of order 5 the elements of order 5 are $\{x, x^2, x^3, x^4\}$
 Thus, H has one of these; however, since $\langle x \rangle = \langle x^2 \rangle = \langle x^3 \rangle = \langle x^4 \rangle$ this implies $\langle x \rangle \subset H$ and since they both have 5 elements this means $H = \langle x \rangle$

Case 4: H has order 10

- If H has order 10, it must be the whole group D_4 .
- Therefore, H would be D_5 , which is already on our list.

In each case, we've shown that any possible subgroup H of D_5 must be one of the subgroups already listed. Therefore, no additional subgroups can exist, and our list of subgroups is complete.

c)

For $n > 3$

$$D_n = \{e, x, x^2, \dots, x^n - 1, y, yx, yx^2, \dots, yx^{n-1}\}$$

$$x = \frac{360^\circ}{n}$$

Claim: For all $1 \leq j < n$ we have $yx^j = x^{n-j}y$.

Proof. Base Case: $m = 1 \implies yx = x^{n-1}y \checkmark$

Inductive Step: Assume for $k \geq 1$ the statement is true

Consider:

$$\begin{aligned} yx^{k+1} &= yx^k x && \text{Apply inductive hypothesis on } yx^k \\ &= x^{n-k} yx \\ &= x^{n-k} x^{n-1} y \\ &= x^n x^{n-(k+1)} y \\ &= x^{n-(k+1)} y \end{aligned}$$

□

This $\implies x^m y = yx^{n-m}$ also

Consider:

$$R = \begin{pmatrix} \cos\left(\frac{360^\circ}{n}\right) & -\sin\left(\frac{360^\circ}{n}\right) \\ \sin\left(\frac{360^\circ}{n}\right) & \cos\left(\frac{360^\circ}{n}\right) \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$SR^j = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ \sin\left(\frac{360^\circ j}{n}\right) & \cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

Let $0 \leq j < n$

$$R^{-j}S = \begin{pmatrix} \cos\left(-\frac{360^\circ j}{n}\right) & -\sin\left(-\frac{360^\circ j}{n}\right) \\ \sin\left(-\frac{360^\circ j}{n}\right) & \cos\left(-\frac{360^\circ j}{n}\right) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Since $\cos(-\theta) = \cos(\theta)$ and $\sin(-\theta) = -\sin(\theta)$, this becomes:

$$R^{-j}S = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & \sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

The matrix multiplication gives:

$$= \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

which is the same as the left-hand side.

Thus, we have shown that:

$$SR^j = R^{-j}S$$

To show $R^{-j} = R^{n-j}$, consider: Let $0 \leq j < n$

$$R^{-j} = \begin{pmatrix} \cos\left(-\frac{360^\circ j}{n}\right) & -\sin\left(-\frac{360^\circ j}{n}\right) \\ \sin\left(-\frac{360^\circ j}{n}\right) & \cos\left(-\frac{360^\circ j}{n}\right) \end{pmatrix}$$

$$R^{n-j} = \begin{pmatrix} \cos\left(\frac{360^\circ(n-j)}{n}\right) & -\sin\left(\frac{360^\circ(n-j)}{n}\right) \\ \sin\left(\frac{360^\circ(n-j)}{n}\right) & \cos\left(\frac{360^\circ(n-j)}{n}\right) \end{pmatrix}$$

Noting that:

$$\cos\left(\frac{360^\circ(n-j)}{n}\right) = \cos\left(\frac{360^\circ n}{n} - \frac{360^\circ j}{n}\right) = \cos\left(-\frac{360^\circ j}{n}\right)$$

$$\sin\left(\frac{360^\circ(n-j)}{n}\right) = \sin\left(\frac{360^\circ n}{n} - \frac{360^\circ j}{n}\right) = -\sin\left(\frac{360^\circ j}{n}\right)$$

we have:

$$R^{n-j} = R^{-j}$$

Subgroup of $GL_2(\mathbb{R})$: $\langle R, S \mid R^n = S^2 = I, SR = R^{n-1}S \rangle = A$

- (Associativity) Given by matrix multiplication
- (Identity)

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

this is a zero-degree rotation

- (Inverses) Each rotation matrix R^k s.t. $1 \leq k < n$ has an inverse R^{n-k} . Each reflection matrix SR^j s.t. $0 \leq j < n$ has an inverse itself.

$$R^k R^{n-k} = R^n = I = R^n = R^{n-k} R^k$$

$$SR^j SR^j = SR^j R^{n-j} S = SIS = SS = I$$

- (Closure)

1. Let R^j and R^k s.t. $1 \leq j, k < n$

$$R^j R^k = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ \sin\left(\frac{360^\circ j}{n}\right) & \cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} \begin{pmatrix} \cos\left(\frac{360^\circ k}{n}\right) & -\sin\left(\frac{360^\circ k}{n}\right) \\ \sin\left(\frac{360^\circ k}{n}\right) & \cos\left(\frac{360^\circ k}{n}\right) \end{pmatrix}$$

Using the angle addition formulas $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$ and $\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$

$$= \begin{pmatrix} \cos\left(\frac{360^\circ j+k}{n}\right) & -\sin\left(\frac{360^\circ j+k}{n}\right) \\ \sin\left(\frac{360^\circ j+k}{n}\right) & \cos\left(\frac{360^\circ j+k}{n}\right) \end{pmatrix}$$

This is another rotation matrix specifically $R^{j+k \mod n}$

2. Let SR^j and R^k be a reflection and a rotation s.t. $0 \leq j < n$ and $1 \leq k < n$. Then

$$(SR^j)R^k = SR^{j+k \mod n}$$

This is another reflection in the group.

$$R^k(SR^j) = (R^k S)R^j = S(R^{n-k}R^j) = SR^{n+j-k \mod n}$$

This is another reflection in the group.

3. Let SR^j and SR^k be reflections s.t. $0 \leq j, k < n$

$$SR^j SR^k = R^{n-j} SSR^k = R^{n+k-j \mod n}$$

This is another rotation in the group.

Let $\varphi : D_n \mapsto A$ s.t. $\varphi(x^j) = R^j$, and $\varphi(yx^j) = SR^j$ s.t. $0 \leq j < n$

Well Defined: Le

Homomorphism: Let $a, b \in D_n$

1. If $a = x^j$ and $b = x^k$ s.t. $1 \leq j, k \leq n$

$$\varphi(x^j x^k) = \varphi(x^{j+k}) = R^{j+k} = R^j R^k = \varphi(x^j) \varphi(x^k)$$

2. If $a = x^j$ and $b = yx^k$ s.t. $0 \leq j, k < n$

$$\varphi(x^j yx^k) = \varphi(yx^{n-j} x^k) = SR^{n+k-j} = R^j SR^k = \varphi(x^j) \varphi(yx^k)$$

$$\varphi(yx^k x^j) = \varphi(yx^{k+j}) = SR^{k+j} = SR^k R^j = \varphi(yx^k) \varphi(x^j)$$

3. If $a = yx^j$ and $b = yx^k$ s.t. $0 \leq j, k < n$

$$\varphi(yx^j yx^k) = \varphi(x^{n-j} y y x^k) = \varphi(x^{n+k-j}) = R^{n+k-j} = R^{n-j} SSR^k = SR^j SR^k = \varphi(yx^j) \varphi(yx^k)$$

Injective: Let $a, b \in D_n$ s.t. $\varphi(a) = \varphi(b)$

Small Proof Before: Let G, H be groups and φ be a homomorphism from G to H then

$$\varphi(g)^{-1} = \varphi(g^{-1}) \forall g \in G$$

Proof. Let $g \in G$

$$e_H \varphi(e_G) = e_H \varphi(e_G e_G) = e_H \varphi(e_G) \varphi(e_G) \implies e_H = \varphi(e_G)$$

$$e_H = \varphi(e_G) = \varphi(gg^{-1}) = \varphi(g) \varphi(g^{-1}) \implies \varphi(g)^{-1} = \varphi(g^{-1}) \quad \square$$

$$\varphi(a) = \varphi(b)$$

$$\varphi(a) \varphi(b)^{-1} = I$$

$$\varphi(ab^{-1}) = I$$

Cases:

1. If $a = x^j$ and $b = x^k$ for $0 \leq j, k < n$

$$I = \varphi(x^j x^{-k}) = R^j R^{-k} \implies j = k$$

Therefore $a = b$

2. If $a = yx^j$ and $b = yx^k$ for $0 \leq j, k < n$

$$I = \varphi(yx^j yx^k) = SR^j SR^k = R^{n-j} SSR^k = R^{n+k-j} \implies k = j$$

Therefore $a = b$

3. If $a = x^j$ and $b = yx^k$ for $0 \leq j, k < n$

$$I = \varphi(x^j y x^k) = R^j S R^k = S R^{n+k-j} \implies R^{j-n-k} = S \Rightarrow \Leftarrow$$

This is a contradiction since this would turn A into only having R

Therefore $a = b$ thus φ is Injective

Surjective:

Let $a \in A$

Cases:

1. If $a = R^k$ s.t. $0 \leq k < n$ Consider $\varphi(x^k) = R^k = a$

2. IF $a = S R^k$ s.t. $0 \leq k < n$ Consider $\varphi(y x^k) = S R^k = a$

Therefore φ is surjective

Therefore φ is bijective

Therefore φ is an isomorphism

d)

Let G be a group Consider $Z(G)$

1. (Identity) Consider $e, g \in G$.
 $eg = ge \implies e \in Z(G)$
2. (Closure) Let $z, h \in Z(G)$ and $g \in G$
 $zhg = zgh = gzh \implies zh \in Z(G)$
3. (Inverses) Let $z \in Z(G)$ and $g \in G$
 $zg = gz \implies g^{-1}z^{-1} = z^{-1}g^{-1} \implies z^{-1} \in Z(G)$

Therefore $Z(G)$ is a subset of G

- If n is odd then $Z(G) = \{e\}$
- If n is even then $Z(G) = \{e, x^{n/2}\}$

Proven in c) $x^j y = y x^{n-j}$ for $1 \leq j < n$

Thus only x^j commutes with y if $j = n - j \implies j = n/2$

and x^j commutes with all x^k since associativity and that $x^k x^j = x^{k+j}$

and since x^j commutes with x^k and y it will commute with all reflections

Therefore this is only possible if n is even.

Geometrically:

When n is odd this is just the do nothing to the n -gon.

When n is even this is just a 180° rotation.

e)

Let G, H be a group and φ be a homomorphism between them and let $g \in G \forall n \in \mathbb{Z}, \varphi(g^n) = [\varphi(g)]^n$

Proof. Base Case: $n = 0 \quad \varphi(g^0) = \varphi(e_1) = e_2 = [\varphi(g)]^0$

Assume statement is true for $k \geq 0$ for $k \in \mathbb{N}$

Consider: $k + 1$

$$\varphi(g^{k+1}) = \varphi(g)\varphi(g^k) = \varphi(g)[\varphi(g)]^k = [\varphi(g)]^{k+1}$$

$$\text{Note: } e_2 = \varphi(e_1) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) \implies [\varphi(g)]^{-1} = \varphi(g^{-1})$$

This proves the statement for all negative integers since we just apply the induction on g then find its inverse □

If φ is an isomorphism then $\text{ord}(g) = \text{ord}(\varphi(g))$

Proof. Assume $g \in G$ s.t. $\text{ord}(g) = n \in \mathbb{N}$

$$\implies \varphi(g^n) = \varphi(e_G) = e_H \iff \text{a proof in c)}$$

$$\implies \varphi(g)^n = e_H \implies \text{ord}(\varphi(g)) \text{ divides } n$$

Since φ is injective, if $\varphi(g)^m = e_H$ for some $m < n$ then $g^m = e_G$ but this contradicts that g has order n

$$\text{Thus } \text{ord}(\varphi(g)) = \text{ord}(g) \quad \square$$

Thus, by the proof above an isomorphism has 2 choices for where x goes either τ or τ^2 and once you pick x there is only one place for x^2

and there is 1 choice for e

and y has 3 choices $\sigma, \sigma\tau, \sigma\tau^2$, and since the isomorphism has to preserve group structure there is only one choice for yx and yx^2 after you pick y

Thus, there are in total 6 isomorphisms.

They are:

- | | | |
|---|---|---|
| 1. • e to e
• x to τ
• x^2 to τ^2
• y to σ
• yx to $\sigma\tau$
• yx^2 to $\sigma\tau^2$ | 3. • e to e
• x to τ^2
• x^2 to τ
• y to $\sigma\tau^2$
• yx to $\sigma\tau$
• yx^2 to σ | 5. • e to e
• x to τ
• x^2 to τ^2
• y to $\sigma\tau^2$
• yx to σ
• yx^2 to $\sigma\tau$ |
| 2. • e to e
• x to τ^2
• x^2 to τ
• y to $\sigma\tau$
• yx to σ
• yx^2 to $\sigma\tau^2$ | 4. • e to e
• x to τ^2
• x^2 to τ
• y to σ
• yx to $\sigma\tau^2$
• yx^2 to $\sigma\tau$ | 6. • e to e
• x to τ
• x^2 to τ^2
• y to $\sigma\tau$
• yx to $\sigma\tau^2$
• yx^2 to σ |

All of these have the properties $\varphi(x)^3 = \varphi(y)^2 = e$ and $\varphi(y)\varphi(x)\varphi(y) = \varphi(x)^2$
 Thus they are homomorphisms and they are obviously bijective by construction thus they are isomorphisms.

Problem 2

Let G be a group.

Consider $(\text{Aut}(G), \text{function composition})$

1. (Associativity) Given by function composition.

2. (Identity) Let $\text{id}(g) = g$ for all $g \in G$

Let φ be another automorphism on G

$$\text{id} \circ \varphi(g) = \varphi(g) = \varphi \circ \text{id}(g)$$

3. (Inverses) Let $\varphi \in \text{Aut}(G)$.

Since φ is a bijection by definition it has an inverse call this φ^{-1}

This inverse is bijective and goes from G to G

Homomorphism:

Let $x, y \in G$ since φ is an automorphism there exists $a, b \in G$ s.t. $x = \varphi(a)$ and $y = \varphi(b)$ thus

$$\varphi^{-1}(xy) = \varphi^{-1}(\varphi(a)\varphi(b))$$

Since φ is a homomorphism

$$\varphi^{-1}(xy) = \varphi^{-1}(\varphi(ab))$$

$$\varphi^{-1}(xy) = ab$$

$$\varphi^{-1}(xy) = \varphi^{-1}(x)\varphi^{-1}(y)$$

Thus φ^{-1} is also an automorphism, so inverses are also in the $(\text{Aut}(G), \text{function composition})$

4. (Closure) Let $\psi, \phi \in \text{Aut}(G)$

• Homomorphism: Let $a, b \in G$

$$\begin{aligned} (\psi \circ \phi)(a \cdot b) &= \psi(\phi(a \cdot b)) \\ &= \psi(\phi(a) \cdot \phi(b)) \end{aligned}$$

(since ϕ is a homomorphism)

$$= \psi(\phi(a)) \cdot \psi(\phi(b))$$

(since ψ is a homomorphism)

$$= (\psi \circ \phi)(a) \cdot (\psi \circ \phi)(b).$$

Therefore, $\psi \circ \phi$ is a homomorphism.

- **Bijjective:**

- **Injective:** Assume $(\psi \circ \phi)(a) = (\psi \circ \phi)(b)$ for some $a, b \in G$. Then:

$$\psi(\phi(a)) = \psi(\phi(b)).$$

Since ψ is injective, we have $\phi(a) = \phi(b)$. Since ϕ is also injective, this implies $a = b$. Therefore, $\psi \circ \phi$ is injective.

- **Surjective:** Let $c \in G$. Since ψ is surjective, there exists $b \in G$ such that $\psi(b) = c$. Since ϕ is surjective, there exists $a \in G$ such that $\phi(a) = b$. Therefore, $(\psi \circ \phi)(a) = \psi(\phi(a)) = \psi(b) = c$, proving that $\psi \circ \phi$ is surjective.

- Since $\psi \circ \phi$ is both a homomorphism and bijective, it is an automorphism of G .

Therefore, $(\text{Aut}(G), \text{function composition})$ is a group.

Separate Proof/Example: Let $G = \langle S \mid R \rangle$ be a group where $S = \{s_1, s_2, \dots, s_n\}$ is a generating set, and R is a set of relations among the generators s.t. a relation in G is an equation of the form:

$$s_1^{k_1} s_2^{k_2} \dots s_n^{k_n} = e \text{ for } k_i \in \mathbb{Z}$$

Let ϕ be an automorphism on G . Let $g \in G$ thus $g = s_1^{k_1} s_2^{k_2} \dots s_n^{k_n}$ for c

$$\phi(g) = \phi(s_1^{k_1} s_2^{k_2} \dots s_n^{k_n}) = \phi(s_1^{k_1}) \phi(s_2^{k_2}) \dots \phi(s_n^{k_n})$$

Thus, knowing $\phi(s_1), \phi(s_2), \dots, \phi(s_n)$ completely determines $\phi(g)$ for any $g \in G$. Hence, the automorphism ϕ is determined by its action on the generators in set S

Let $s_1^{k_1} s_2^{k_2} \dots s_n^{k_n} = e$ be a relation in R

$$\phi(s_1^{k_1} s_2^{k_2} \dots s_n^{k_n}) = \phi(s_1)^{k_1} \phi(s_2)^{k_2} \dots \phi(s_n)^{k_n} = e$$

This shows that the image of a relation under an automorphism is still a valid relation.

Therefore, automorphism preserve the relations among the generators of the group.

Therefore, an automorphism is determined by how it maps the generators of the group, and it preserves the relations among these generators.

a)

Let $C_5 = \langle g \rangle$ where g is a generator. By the proof/exercise above any automorphism $\varphi \in \text{Aut}(C_5)$ is determined by where it sends g and if it preserves relations the only relation in this case is the order of g thus $\varphi(g)$ must be another element with order 5.

In C_5 there are 4 elements with order 5 thus,

$$\varphi_k(g) = g^k, \quad \text{for } k = 1, 2, 3, 4.$$

Therefore, $\text{Aut}(C_5)$ has 4 elements.

Consider φ_2

$$\begin{aligned}\varphi_2^1(g) &= g^2 \\ \varphi_2^2(g) &= \varphi_2(g^2) = g^4 \\ \varphi_2^3(g) &= \varphi_2(\varphi(g)^2) = \varphi_2(g^4) = g^8 = g^3 \\ \varphi_2^4(g) &= \varphi_2(\varphi_2^3(g)) = \varphi_2(g^3) = g\end{aligned}$$

Therefore $\text{Aut}(C_5)$ has an element of order 4 and since the group also has 4 elements proven in class $\text{Aut}(C_5) \cong C_4$

b)

Let $C_8 = \langle g \rangle$ where g is a generator. By the proof/exercise above any automorphism $\varphi \in \text{Aut}(C_8)$ is determined by where it sends g and if it preserves relations the only relation in this case is the order of g thus $\varphi(g)$ must be another element with order 8.

In C_8 there are 4 elements with order 8 thus,

$$\varphi_k(g) = g^k, \quad \text{for } k = 1, 3, 5, 7.$$

Therefore, $\text{Aut}(C_8)$ has 4 elements.

Consider φ_3

$$\begin{aligned}\varphi_3(g) &= g^3 \\ \varphi_3^2(g) &= \varphi_3(g^3) = g\end{aligned}$$

Consider φ_5

$$\begin{aligned}\varphi_5(g) &= g^5 \\ \varphi_5^2(g) &= \varphi_3(g^5) = g\end{aligned}$$

Consider φ_7

$$\begin{aligned}\varphi_7(g) &= g^7 \\ \varphi_7^2(g) &= \varphi_3(g^7) = g\end{aligned}$$

Therefore all non identity elements have order 2 so this group is not cyclic thus proven in class
 $\text{Aut}(C_8) \cong C_2 \times C_2$

c)

Let $C_{11} = \langle g \rangle$ where g is a generator. By the proof/exercise above any automorphism $\varphi \in \text{Aut}(C_{11})$ is determined by where it sends g and if it preserves relations the only relation in this case is the order of g thus $\varphi(g)$ must be another element with order 11.

In C_{11} there are 10 elements with order 11 thus,

$$\varphi_k(g) = g^k, \quad \text{for } k = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.$$

Therefore, $\text{Aut}(C_5)$ has 10 elements.

Consider φ_2

$$\begin{aligned}
 \varphi_2^1(g) &= g^2 \\
 \varphi_2^2(g) &= \varphi_2(g^2) = g^4 \\
 \varphi_2^3(g) &= \varphi_2(\varphi_2^2(g)) = \varphi_2(g^4) = g^8 \\
 \varphi_2^4(g) &= \varphi_2(\varphi_2^3(g)) = \varphi_2(g^8) = g^5 \\
 \varphi_2^5(g) &= \varphi_2(\varphi_2^4(g)) = \varphi_2(g^5) = g^{10} \\
 \varphi_2^6(g) &= \varphi_2(\varphi_2^5(g)) = \varphi_2(g^{10}) = g^9 \\
 \varphi_2^7(g) &= \varphi_2(\varphi_2^6(g)) = \varphi_2(g^9) = g^7 \\
 \varphi_2^8(g) &= \varphi_2(\varphi_2^7(g)) = \varphi_2(g^7) = g^3 \\
 \varphi_2^9(g) &= \varphi_2(\varphi_2^8(g)) = \varphi_2(g^3) = g^6 \\
 \varphi_2^{10}(g) &= \varphi_2(\varphi_2^9(g)) = \varphi_2(g^6) = g
 \end{aligned}$$

Therefore $\text{Aut}(C_5)$ has an element of order 10 and since the group also has 10 elements proven in class $\text{Aut}(C_5) \cong C_{10}$

d)

Let $S_3 = \langle x = (123), y = (12) \mid x^3 = y^2 = e, yx = x^2y \rangle$ where x, y are generators. By the proof/exercise above any automorphism $\varphi \in \text{Aut}(S_3)$ is determined by where it sends x, y and if it preserves relations in this case is the order of x, y thus $\varphi(x)$ has order 3, $\varphi(y)$ has order 2, and $\varphi(y)\varphi(x)\varphi(y) = \varphi(x)^2$.

Given this we have the following possibilities for φ ,

$$\varphi(y) = y, \quad \varphi(x) = x \text{ or } x^2.$$

$$\varphi(y) = yx^2, \quad \varphi(x) = x \text{ or } x^2.$$

$$\varphi(y) = yx, \quad \varphi(x) = x \text{ or } x^2.$$

Thus, there are 6 automorphisms. Therefore, $\text{Aut}(S_3)$ has 6 elements.

1. φ_1 : Identity automorphism

$$\varphi_1(y) = y$$

$$\varphi_1(x) = x$$

Order: 1 (identity map).

2. φ_2 : Cycle the transpositions $y \rightarrow yx^2 \rightarrow yxy$

$$\varphi_2(y) = yx^2$$

$$\varphi_2(x) = x$$

Order: 3 (Applying φ_2 three times gives the identity automorphism).

3. φ_3 : Swap transpositions y and yx^2 , keeping yx fixed

$$\varphi_3(y) = yx^2$$

$$\varphi_3(x) = x^2$$

Order: 2 (Applying φ_3 twice gives the identity automorphism).

4. φ_4 : Cycle the transpositions $y \rightarrow yx \rightarrow yx^2$

$$\varphi_4(y) = yx$$

$$\varphi_4(x) = x$$

Order: 3 (Applying φ_4 three times gives the identity automorphism).

5. φ_5 : Swap transpositions y and yx , keeping yx^2 fixed

$$\varphi_5(y) = yx$$

$$\varphi_5(x) = x^2$$

Order: 2 (Applying φ_5 twice gives the identity automorphism).

6. φ_6 : Identity on y and swapping the 3-cycles x and x^2

$$\varphi_6(y) = y$$

$$\varphi_6(x) = x^2$$

Order: 2 (Applying φ_6 twice gives the identity automorphism).

Therefore, $\text{Aut}(S_3)$ has no elements of order 6, so it is not cyclic. Thus, proven in class since it has 6 elements $\text{Aut}(S_3) \cong S_3$

e)

Let $C_2 \times C_2 = \{e, x = (1, 0), y = (0, 1), (1, 1)\}$. By the proof/exercise above any automorphism $\varphi \in \text{Aut}(C_2 \times C_2)$ is determined by where it sends x, y and if it preserves relations in this case is the order of x, y thus $\varphi(x)$ and $\varphi(y)$ have order 2, and $\varphi(y)\varphi(x)\varphi(y) = \varphi(x)$.

Given this we have the following possibilities for φ ,

$$\varphi(y) = y, \quad \varphi(x) = x \text{ or } xy.$$

$$\varphi(y) = x, \quad \varphi(x) = y \text{ or } xy.$$

$$\varphi(y) = xy, \quad \varphi(x) = x \text{ or } y.$$

Thus, there are 6 automorphisms. Therefore, $\text{Aut}(C_2 \times C_2)$ has 6 elements.

1. φ_1 : Identity automorphism

$$\varphi_1(y) = y$$

$$\varphi_1(x) = x$$

Order: 1 (identity map).

2. φ_2 :

$$\varphi_2(y) = y$$

$$\varphi_2(x) = xy$$

Order: 2 (Applying φ_2 twice gives the identity automorphism).

3. φ_3 : Swap transpositions y and yx^2 , keeping yx fixed

$$\varphi_3(y) = x$$

$$\varphi_3(x) = y$$

Order: 2 (Applying φ_3 twice gives the identity automorphism).

4. φ_4 :

$$\varphi_4(y) = x$$

$$\varphi_4(x) = xy$$

Order: 3 (Applying φ_4 three times gives the identity automorphism).

5. φ_5 :

$$\varphi_5(y) = xy$$

$$\varphi_5(x) = x$$

Order: 2 (Applying φ_5 twice gives the identity automorphism).

6. φ_6 :

$$\varphi_6(y) = xy$$

$$\varphi_6(x) = y$$

Order: 3 (Applying φ_6 three gives the identity automorphism).

Therefore, $\text{Aut}(C_2 \times C_2)$ has no elements of order 6, so it is not cyclic. Thus, proven in class since it has 6 elements $\text{Aut}(C_2 \times C_2) \cong S_3$

f)

By the proof/exercise above any automorphism $\varphi \in \text{Aut}(\mathbb{Z})$ is determined by where it sends 1 thus $\varphi(1)$ must be another generator of $\mathbb{Z}$.

In $\mathbb{Z}$ there are 2 generators,

$$\varphi_k(1) = (-1)^k, \quad \text{for } k = 0, 1.$$

Therefore, $\text{Aut}(\mathbb{Z})$ has 2 elements.

Thus, proven in class groups with an order equal to a prime number are isomorphic to C_p where p is prime thus $\text{Aut}(\mathbb{Z}) \cong C_2$

Problem 3

Proof. Let p be prime and G be a group of order $2p$

1. Assume $p = 2$

Thus G has 4 elements

Thus proven in class G is either isomorphic to C_4 or $C_2 \times C_2$

(a) If G is cyclic then it is isomorphic to C_4

(b) If G is not cyclic it is isomorphic to $C_2 \times C_2$ Since D_2 is not cyclic either this means $C_2 \times C_2 \cong D_2$ thus $D_2 \cong G$

2. Assume G has an element g s.t. $\text{ord}(g) = 2p$ for odd primes

$\implies \langle g \rangle$ has $2p$ elements since G also has $2p$ elements and $\langle g \rangle \subseteq G$ thus there is no element in G that is not in $\langle g \rangle$ thus $G = \langle g \rangle$ thus proven in class $G \cong C_{2p}$

3. Assume G has no elements of order $2p$ and p is odd

For contradiction assume G has no elements of order $p \implies$ by my proof above that all non-identity elements have order of 2

Take $e \neq a \neq b \neq e \in G$

Consider $\{e, a, b, ab\} \leq G \implies$ by Lagrange's Theorem that 4 divides $2p \implies 2$ divides $p \Rightarrow \Leftarrow$

Thus G has an element $h \in G$ with order p . Take $\langle h \rangle = H$ and proven above H is a subgroup of G . By Lagrange's Theorem $[G : H] = \frac{|G|}{|H|} = \frac{2p}{p} = 2$. Take $g \notin H \implies G = H \sqcup gH$

Consider: g^2H

- (a) Assume $g^2H = H \implies g^2 \in H$

$\implies \text{ord}(g^2)$ divides p by Lagrange's Theorem $\implies \text{ord}(g^2) = 1$ or $\text{ord}(g^2) = p$

- i. If $\text{ord}(g^2) = 1 \implies g^2 = e \implies \text{ord}(g)$ divides 2 $\implies \text{ord}(g) = 2$ or 1

A. If $\text{ord}(g) = 1$ then $\langle g \rangle = \langle g^2 \rangle = \{e\}$ which is a subset of H which $\implies g \in H$ this is a contradiction

- ii. If $\text{ord}(g^2) = p \implies g^{2p} = e \implies \text{ord}(g)$ divides $2p$ and since there is no elements of order $2p$

$\implies \text{ord}(g) = 1$ or p

A. If $\text{ord}(g) = p$ then $\langle g \rangle = \langle g^2 \rangle$ which is a subset of H which $\implies g \in H$ this is a contradiction

B. If $\text{ord}(g) = 1$ then $g = e$ but $g^2 \neq e$ since $\text{ord}(g^2) = p$ this is a contradiction

Therefore $\text{ord}(g) = 2$ by 3.(a).i

$$(b) \ g^2H = gH$$

Take $h, j \in H$ s.t. $g^2h = gj \implies g = jh^{-1} \in H$ this is a contradiction

Therefore anything not in H has order 2.

$$gh \in gH \implies gh \notin H \implies \text{ord}(gh) = 2$$

$e = (gh)^2 = ghgh \implies h^{-1}g^{-1} = gh \implies gh = h^{p-1}g \implies gh^k = h^{p-k}g$ Since Cosets are a partition

$G = H \sqcup gH = \{e, h, h^2, \dots, h^{p-1}, g, gh, \dots, gh^{p-1}\} = \langle h, g \mid e = h^p = g^2, gh = h^{p-1}g \rangle \cong D_p$
by similar proof in problem 1 c).

□

Problem 4

Proof. ($\implies$) Assume HK is a subgroup of G

Let $j \in HK$ this means there exists $h \in H$ and $k \in K$ s.t. $j = hk$

Consider j^{-1} since HK is a subgroup of G this implies $j^{-1} \in HK$

Consider $j^{-1} = k^{-1}h^{-1}$ and since H and K are subgroups of G this implies that $k^{-1} \in K$ and $h^{-1} \in H$ thus $j^{-1} \in KH$ thus $HK \subseteq KH$

Let $j \in KH$ this means there exists $h \in H$ and $k \in K$ s.t. $j = kh$

Consider $j^{-1} = h^{-1}k^{-1}$ and since H and K are subgroups of G this implies that $k^{-1} \in K$ and $h^{-1} \in H$ thus $j^{-1} \in HK$ and since HK is a subgroup of G this implies that $j \in HK$ thus $KH \subseteq HK$

Therefore $HK = KH$ □

Proof. ($\impliedby$) Assume $HK = KH$

- (Identity) Since H and K are subgroups $e \in H \cap K$ thus $ee = e \in HK$
- (Inverses) Let $j \in HK \implies \exists h \in H$ and $k \in K$ s.t. $j = hk$
 Consider $j^{-1} = k^{-1}h^{-1}$ since H and K are subgroups $k^{-1} \in K$ and $h^{-1} \in H$
 therefore $j^{-1} \in KH = HK$
- (Closure) Let $j, i \in HK = KH \implies \exists h, h' \in H$ and $k, k' \in K$ s.t. $j = hk$ and $i = h'k'$
 $ji = hkh'k'$ since $KH = HK$ there exists an element n in HK s.t. $kh' = n$
 $ji = hnk'$ since $n \in HK \implies \exists s \in H$ and $t \in K$ s.t. $n = st$
 $ji = hstk'$ since H is a subgroup $hs \in H$ and since K is a subgroup $tk' \in K$ thus $ij \in HK$

Therefore HK is a subgroup of G . □

Problem 5

Proof. Initially, consider the set of all pairs (h, k) where $h \in H$ and $k \in K$. The total number of such pairs is $|H| \times |K|$. If all such pairs (h, k) produced distinct elements in HK , then $|HK| = |H| \times |K|$. However, since H and K may have non-trivial intersection, some products hk may coincide, leading to overcounting.

Suppose $h_a k_a = h_b k_b$ for some $h_a, h_b \in H$ and $k_a, k_b \in K$. This implies:

$$h_a^{-1} h_b = k_a k_b^{-1}.$$

The left-hand side, $h_a^{-1} h_b$, is an element of H , and the right-hand side, $k_a k_b^{-1}$, is an element of K . Thus, $h_a^{-1} h_b$ must be an element of $H \cap K$, meaning:

$$h_a^{-1} h_b \in H \cap K \quad \text{and} \quad k_a k_b^{-1} \in H \cap K.$$

Therefore, if $h_a k_a = h_b k_b$, then the pairs (h_a, k_a) and (h_b, k_b) correspond to the same element in HK , and the number of distinct elements in HK is reduced by a factor equal to the size of the intersection $H \cap K$.

Therefore, the number of distinct elements in HK is exactly $\frac{|H| \cdot |K|}{|H \cap K|}$, establishing that

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

□